

COMPREHENSIVE PROFILING OF ALTERNATIVE PROMOTER EVENTS IN DISEASE BIOLOGY

Internship Report

by

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Rishita Jain

Abstract

Alternative promoter usage represents a very important mechanism which is responsible for transcriptomic diversity because it allows generating various RNA transcripts from the same gene. Despite the strong correlation between abnormal TSS usage and conditions such as heart failure or cancer, the accurate measurement of promoter activity with the help of RNA short-read data is quite problematic. It stems from the problems related to low efficiency of transcript end mapping as well as the variability in different software algorithms. In this study, the SnakeAltPromoter algorithm was used to assess the performance of three frequently used algorithms, namely, proActiv, Salmon, and DEXSeq, for estimating promoter activity. As a starting point, the GSE147236 dataset (heart failure) was selected in order to test these algorithms in terms of sensitivity and specificity. The pipeline was validated by applying it to HCC data for an exploration of promoter dynamics relevant to the disease state. The results from this experiment reveal the link between transcription and epigenetics and the discovery of new alternative promoters. Some genes like CD63, TNS3, SETBP1 were identified as genes with alternative promoter usage in the HCC dataset. These genes were validated by using existing literature which in turn validated the pipeline that was used. This study sheds light on the significance of analyzing promoter activity apart from regular gene expression analysis. This is important for discovering promoter regions as biomarkers or therapeutic targets so that these can be altered to treat various diseases.

Introduction

The fundamental basis of molecular biology is the linear flow of information that begins with DNA, proceeds through RNA, and ultimately results in proteins. Initially, it was thought that there was a straightforward relationship wherein a specific gene would produce a specific protein. With the emergence of next-generation sequencing technologies, it has become clear that the transcriptome organization is extremely complicated and diverse. One of the major contributors to such diversity, aside from alternative splicing, is the employment of Alternative Promoters (APs). APs represent unique regulatory sequences upstream or in introns of a gene that autonomously drive its transcription (Ayoubi et al., 1996). The ability to employ multiple transcription start sites allows a single gene to produce a variety of mRNA transcripts. These usually differ in 5' UTRs, which are involved in translation regulation, stability of mRNAs, and their sub-cellular transport. AP activity often produces altered or novel protein isoforms with different cellular functions (Reyes et al., 2017).

The ability to modulate APs is necessary for biological phenomena like embryo development, cell differentiation, and reactions to stressful conditions. Therefore, the dysfunction of this process is closely connected with the complex disorders in humans. When referring to cardiomyopathy, including heart failure, the heart muscle experiences substantial alterations both structurally and metabolically. The process is not limited to changes in gene expression but includes more delicate modifications in RNA isoforms. In oncology, one of the most common strategies used to alter the transcriptome is the manipulation of APs. Cancer cells exploit this network to silence or activate genes that function as tumor suppressors or oncogenes without the need for genetic mutations. In spite of its importance to biology, quantifying changes in APs is an unresolved issue.

HCC is the most common form of liver cancer and accounts for most cases of liver cancer diagnosed today. Liver cancer has now become one of the major killers among the various cancers. This is largely attributed to the fact that HCC is very hard to diagnose early, and patients do not exhibit symptoms until the tumor is well-established, whereupon it becomes almost impossible to treat using the regular methods. Thus, there is an urgent need in the medical field to identify unique molecular markers that will assist in diagnosing the disease much earlier or in predicting its aggressive nature. (Zeng et al., 2020, Mauro et al., 2025, Jin et al., 2019)

The identification and measurement of alternative promoter use from short-read RNA-seq pose significant computational challenges. In conventional RNA-seq, the fragmentation of mRNAs

causes short sequences whose positions cannot be unambiguously determined. Short sequences mapping to the most 5' end of transcripts cause ambiguities particularly at those loci whose transcripts overlap each other. Although methods such as CAGE and long-read sequencing can identify TSSs directly, they are expensive compared to RNA-seq (Kodzius et al., 2006).

At present, scientists use methods such as Salmon, DEXSeq, and proActiv for estimating promoter counts using bulk RNA-seq data sets. Salmon uses a quasi-mapping method for analyzing transcript fragments, while DEXSeq assesses exon bin usage, and proActiv makes use of spliced read counts as indicators of promoter function. This difference in mathematical models results in different results, which creates a crucial obstacle for the further development of biological research. It is important to find out which tool is more reliable during which situation. The tool proActiv works well with intron-less transcripts while Salmon is a better choice for transcripts that contain introns. DEXSeq is useful for determining the exon architecture in the transcripts. To make it easy to compare these three tools at the same time, SnakeAltPromoter, an automated, scalable pipeline was designed. It takes fastq files as input and automatically runs the entire workflow (Tan et al., 2026).

This thesis aims to compare the tools proActiv, Salmon and DEXSeq using the SnakeAltPromoter pipeline on a heart failure RNA-seq dataset to compare their sensitivity, limitations and accuracy. This pipeline is validated using a Hepatocellular Carcinoma dataset to find possible biomarkers in the form of alternative promoters.

Review of Literature

According to new transcriptomic information, the conventional gene regulation paradigm has undergone a complete overhaul, since it has become clear that the mammalian genome makes extensive use of alternate promoters in order to fully exploit its coding capacity. Genomic evidence suggests that more than fifty percent of all protein coding genes are regulated by different promoters. The alternate promoters are not mere insurance mechanisms but are distinct switches which operate under specific conditions through the activity of distinct transcription factors and epigenetic alterations. Promoter switching plays a crucial role during organogenesis and cell differentiation, and the selection of an alternate promoter results in the incorporation of different sets of 5' untranslated regions (Sun et al., 2025)

However, when the alternative transcription initiation site is positioned distal to the canonical translation initiation site, the produced mRNA produces a completely new protein isoform. These alternative isoforms typically fail to contain vital amino-terminal domains. Proteins'

ability to interact with other molecules depends on how their surfaces are shaped. When these regions are missing, the cellular signaling circuitry has to be completely reconfigured, which changes how enzymes work, where proteins are located, and how complexes are formed. Pre-transcriptional control is a fast way for cells to change their functional characteristics without changing their DNA, making it an important part of adaptive and oncogenic evolution (Gacita et al., 2020).

Traditionally, the accurate determination of transcription start sites has been achieved through methods like Cap Analysis of Gene Expression that employ physical sequencing approaches. Using a unique chemical procedure that enables the isolation of only the 5' cap of newly transcribed eukaryotic RNAs, this approach offers high-resolution promoter profiling at the single-nucleotide level. The capabilities of this approach extend beyond known genes, enabling the measurement of bi-directionally transcribed, low abundance RNA enhancers, hence mapping the entire regulation pattern of the tissue under study. Unfortunately, the complexity of library preparation, large sequencing depth, and the associated costs make this approach unfeasible for clinical studies and cohort studies (Tan et al., 2026). Since RNA sequencing represents the entire fragmented transcriptome and is extremely economical, it has been adopted as the method of choice for detecting differential gene expressions (Ozsolak et al., 2010). However, the key issue associated with RNA sequencing is the computational reconstruction of the fragmented transcripts into a continuous sequence, which is extremely prone to various types of biases based on sequence context, fragmentation procedures, and even complicated splicing events. Traditional computational methods sidestepped these issues and determined an expression value for the entire gene based on the summation of all its reads. This practice is extremely inaccurate and problematic when one is interested in promoter dynamics, since a reduction in the activity of a major promoter due to a certain disease while the activation of a cryptic promoter leads to no overall change in the gene reads (Tan et al., 2025).

ProActiv's biological assumption is that, in the case of virtually all gene transcription initiation processes, RNA splicing occurs almost instantaneously after the process of transcription initiation, occurring in the first intron. ProActiv quantifies the exact reads occurring within the splice junctions produced by advanced alignment tools like STAR, and this splice junction quantification acts as an extremely reliable metric for the activity of the immediately preceding promoter. Although this method has extremely high accuracy levels and can be normalized with DESeq2 size factors, it has limitations since it only works for genes with introns (Zhang et al., 2024).

Salmon applies a two-step parallel inference method along with sophisticated bias models which account for biases in fragment length distribution and GC-content. Contrary to conventional base-to-base alignment strategy, Salmon aligns the reads against the index-transcriptome, making the quantification process extremely efficient. In the case of alternate promoter studies, customized computational codes sum up the abundance estimates of all the transcripts generated by the identical promoter. This approach is extremely reliable and works remarkably well for loci with fewer introns and in the case of low-depth sequencing (Patro et al., 2017).

DEXSeq works on the principle that overlapping structures of genes are flattened out through mathematical calculations. In doing so, DEXSeq allows for the identification of reads that are mapped only to the unique bins of first exon, which is linked to different promoters. The use of DEXSeq also involves statistical tests based on generalized linear models, which are used to estimate the dispersion of the biological model using the negative binomial distribution (Anders et al., 2012).

A study explored the use of alternative promoter usage in pan-cancer transcriptomes, trying to find out how promoter selection influences cancers. It is important to point out the goal of inferring promoter activity based on RNA-seq data and then analyzing its influence on gene regulation and outcome. The study involved RNA-seq data of 18,468 samples of 42 cancer types (from TCGA and normal tissue samples). For estimating promoter activity, the authors used available tools (presumably including ProActiv). They studied differential promoter selection in tissue type, tumor type, and patient comparison. Survival analysis was performed comparing promoter-level versus gene-level predictions. As a result, the authors found that "alternative promoters are a major contributor to context-specific regulation". It appears that many promoters are differentially regulated in cancer as opposed to normal tissues and subtypes. In addition, promoters gave better survival prediction results than genes (Demircioğlu et al., 2019).

Another study explored alternative promoter utilization in TNBC, with an emphasis on subtype-specific regulation and clinical implications. Here, the goal was to identify tumor-specific Active Alternative Promoters (AAPs) in TNBC and evaluate their impact on patients' prognosis. Using RNA-seq data from 360 TNBC tumors and 88 paired normal breast tissue specimens, the authors measured promoter activity (using RNA-seq data) for each sample. Subsequently, they identified promoters showing differential activity between tumors and normals (TNBC-specific AAPs), as well as within TNBC subtypes. In addition, ChIP-seq data for H3K4me3 and H3K27ac histone modifications was used for validation. Kaplan-Meier

analyses of TNBC subtypes were followed by Cox proportional hazard regression, with a multivariate approach involving promoter activity and clinicopathological characteristics. Several examples of TNBC-specific AAPs could be cited in this study, such as a promoter switch in HDAC9, with a decrease in promoter pr1077 activity and increased pr1079 activity (associated with differential H3K4me3/H3K27ac histone marks). Examples of subtype-specific AAPs include upregulated SEC31A (in basal) and downregulated AKAP9. HUWE1 and FTX alternative promoters were independently associated with relapse-free survival, and their incorporation into the model with tumor size and Ki67 resulted in a prognostic AUROC score of 0.73 (with NRI ~0.4). (Jit et al., 2025)

Knowing that any one algorithm could not be error-free, the newly designed SnakeAltPromoter process merges all three approaches within a very scalable Snakemake pipeline. Combining annotations of the genome together with parallel processing, the tool makes it possible to create a benchmarking platform to identify the hidden world of the transcriptome in massive public databases (Tan et al., 2026).

In the landmark research by Gacita et al. (2020) on the GSE147236 dataset, about 23,000 active promoters and 5,000 active enhancers were identified in the human left ventricle, proving that close to twenty percent of cardiac genes are controlled by more than one active promoter. In the case of heart failure, there are many instances of alternative promoter switching leading to changes in the N-terminus of essential cardiac proteins. One such case is the PRKAG2 gene, which produces the non-catalytic gamma-2 isoform of AMP-activated protein kinase. The function of the AMP-activated protein kinase is to serve as a cellular energy sensor. Heart failure causes the myocardium to undergo an irreversible shift from the conventional promoter gamma-2b to the alternative promoter gamma-2-3b. The reason for this is that both the promoters produce distinct amino termini. Therefore, this process leads to a change in the distribution and activity of the kinase molecule, resulting in the development of glycogen excess, hypertrophy, and potentially fatal arrhythmia (Gacita et al., 2020).

Materials and Methods

DATA COLLECTION

Paired-end bulk RNA-seq fastq files from the datasets with GSE ID's GSE147236 and GSE124535 were downloaded from the Gene Expression Omnibus (GEO) database using SRA Toolkit(v3.2.1). GSE147236 comprises of 7 left-ventricle samples (3 healthy and 4 heart failure). GSE124535 comprises of 35 non-tumor and 35 HCC samples. For genomic alignment

and annotation, the GENCODE comprehensive annotation (hg38, v44 and v46) GTF and primary FASTA files were downloaded (Harrow et al.,2012).

COMPUTATIONAL INFRASTRUCTURE

With the high amount of sequencing data, the processing needed to be performed using a High-Performance Computing cluster, which used SLURM for scheduling the tasks. For full computational reproducibility, a private computing environment had to be set up. The Conda environment, known as SnakeAltPromoter, was initialized for Python version 3.13, which included the channels conda-forge, bioconda, and defaults. After the activation of the environment, the main pipeline was deployed through the Python package manager pip with the flag "--no-cache-dir" to install it on the local machine.

The system check was successful with the retrieval of the help guide of the software. Scripts were written for connecting the SLURM job scheduler, utilizing 40 CPU cores and 200 GB of memory.(Tan et al., 2026)

PIPELINE ARCHITECTURE AND EXECUTION

The SnakeAltPromoter analysis pipeline was implemented to process the RNA-seq data sets obtained. It involves setting up the genome index using the input gene annotation and fasta files. It is done via a command known as Genomesetup as we can see in the code below:

```
Genomesetup --organism hg38 --organism_fasta "$(pwd)/snakealtpromoter/data/hg38.fa" -  
-genes_gtf "$(pwd)/snakealtpromoter/data/hg38.gtf" -o ./genome --threads 30
```

Once that is done, the pipeline features an effective quality control system built into it. The command Snakealtpromoter is used to run the rest of the pipeline as we can see in the code below:

```
Snakealtpromoter -i "$(pwd)/snakealtpromoter/data/heartdata" --genome_dir  
"$(pwd)/genome" -o heartrnaseq_output2 --threads 30 --organism hg38 --trim --  
sample_sheet "$(pwd)/snakealtpromoter/data/samplesheet/Heart_RNAseq.tsv"
```

First, FastQC (v0.12.1) analyzes the quality of the reads and identifies potential artifacts in the sequencing process. Following that, TrimGalore (v0.6.10) is activated via the '--trim' option to trim poor-quality reads and adapter sequences from the raw data. Post-QA, the pipeline undergoes a highly optimized two-pass alignment process using the STAR (v2.7.11b)

algorithm, resulting in the production of sorted BAM files and splicing junction expression matrices (Dobin et al., 2012). Parallel quantification is carried out for each of the main mathematical models – the proActiv (v1.16.0) method quantifies junctions that span the first intron, Salmon (v1.10.3) quantifies total transcripts from index based quasi-mapping, and DEXSeq (v1.52.0) quantifies coverage of first exons into counting bins. (Tan et al., 2026)

PROMOTER IDENTIFICATION AND PROMOTER ACTIVITY QUANTIFICATION

Promoter-by-sample matrices obtained from the previous step were normalized, and promoters that demonstrated a median absolute activity less than the threshold of 0.25 were automatically removed in order to remove noise from the subsequent analyses. Internal promoters (those which emerged solely within the internal coding exon sites) were also omitted from certain statistical groups.

The counts were subjected to log2 transformation, while the promoters were deterministically grouped into major, minor/alternative, or inactive, based on their expression relative to others within a particular genetic locus. The DEGs and DRPs were identified through the use of the edgeR and limma-voom libraries to normalize raw matrices and obtain the Counts Per Million. A Wilcoxon rank-sum test was consistently carried out for the filtered genome to determine any changes in distributions for the two groups. The calculated p-values were corrected for multiple hypotheses testing using the FDR approach. The strict cut-off criteria used for DEGs and DRPs include having a p-value less than 0.05 along with an absolute log2 fold change greater than 1.0.(Dong et al., 2022, Jit et al., 2025)

ACTIVE ALTERNATIVE PROMOTER DETECTION

We measured gene expression along with absolute promoter activity to identify AAPs. We filtered the genes by only taking genes with two or more promoters. Non-significant (p value > 0.05) genes having significant upregulated and downregulated DRPs in terms of absolute promoter activity and median absolute promoter activity greater than 0.25 were taken for further investigation to identify AAPs.

DOWNSTREAM ANALYSIS

After quantification of the promoters' activity, SnakeAltPromoter performed a downstream analysis to make biological sense out of the resulting numeric matrices. All visualizations and

plotting were done using the R environment with the use of the ggplot2 library. All promoters, as well as their corresponding genes, were strictly grouped into three categories- “Major,” “Minor,” and “Inactive.” Stacked histograms were produced to visualize the overall representation of these categories in the data set. To determine the proportion of promoters with respect to promoter multiplicity, a bar plot was created to display the proportion of active promoters among both single-promoter and multi-promoter genes. In addition, to see how promoter activity is linked to gene expression, a scatter plot was generated to show the dependence between the average expression of genes (the y axis) and the activity of major promoters (the x axis).

In the next part of downstream analysis, focus was put on the genes showing transcriptomic rewiring in particular. These were the non-significant genes having Differential Regulatory Promoters (DRPs). For the selected genes, box plots were produced for both gene expression levels as well as absolute promoter activities under the two contrasting biological conditions (such as healthy and diseased state).

SURVIVAL ANALYSIS

To determine the biological significance of the transcriptomic alterations revealed through the pipeline, the survival analysis was performed using the HCC dataset. Clinical annotation including Disease-Free Survival (DFS) time points in months and recurrence events was obtained along with the RNA-sequencing information available in the GSE124535 dataset.

For some selected DRPs of interest, especially the ones in genes showing overall expression decoupling, patients were classified into high and low expression groups according to the absolute promoter activity level in each group.

The prognostic stratification was performed according to the Kaplan-Meier approach to estimate probabilities of disease-free survival as time progresses. The significance of the differences in survival probability between patients with high promoter activity compared to those with low promoter activity was evaluated by performing a log-rank test. A p-value from the log-rank test of less than 0.05 was defined as significant enough to prove that there exists a predictive association between the use of a particular transcription start site and the prognosis of the patients. All survival analyses and visualizations (Kaplan-Meier plots) were done using the R programming language.

CHIP-SEQ DATA COLLECTION AND ANALYSIS

The hg38 aligned bigwig files for HCC cell line PLC were retrieved from ChIP-Atlas tool to validate H3K4me3 and H3K27ac signals. Track plots were generated using IGV tool by uploading the bigwig files and the bed file containing information about the transcripts for the gene SETBP1.

Results and Discussion

BENCHMARKING OF COMPUTATIONAL TOOLS USING SNAKEALTPROMOTER PIPELINE

Evaluating the interpretation of the promoter region by different quantification methods was one of the major goals of this study.

DATASET 1 (GSE147236):

Applying the SnakeAltPromoter pipeline to the heart failure gene expression data set (GSE147236), we discovered 90,164 promoters in the hg38-v46 genome. The effectiveness of three integrative tools, proActiv, Salmon, and DEXSeq, was assessed.

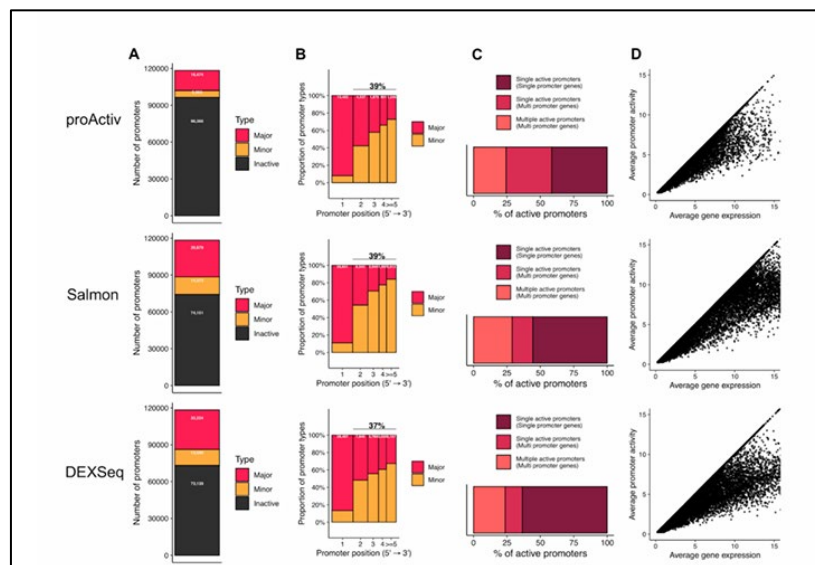


Figure 1. Taken from Tan et al., 2026. A) Classification of promoters into Major, Minor and Inactive promoters. B) Distribution of major and minor promoters across Transcription start sites (TSSs). C) Classification of promoters into single active promoters (single promoter and multi promoter genes) and multi active promoters (multi promoter genes). D) Correlation of major promoter activity and gene expression for all major promoters.

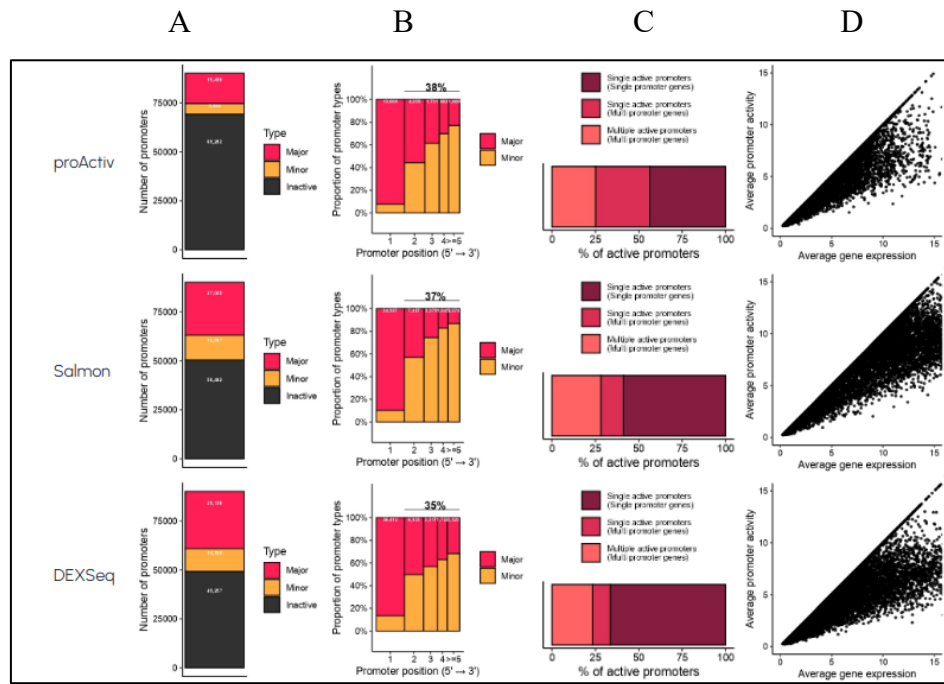


Figure 2. Comparison of tools in GSE147236 A) Classification of promoters into Major, Minor and Inactive promoters. B) Distribution of major and minor promoters across Transcription start sites (TSSs). C) Classification of promoters into single active promoters (single promoter and multi promoter genes) and multi active promoters (multi promoter genes). D) Correlation of major promoter activity and gene expression for all major promoters.

According to our results (Figure 4.2), proActiv recognizes less Major promoters (15,498) than Salmon (27,025) and DEXSeq (29,198). This may indicate that the junction-based approaches are more sensitive to identifying confident start sites. At the same time, Salmon and DEXSeq detected almost double the number of Minor promoters (12,657 and 11,709 respectively) compared to proActiv (5,384). It proves the ability of DEXSeq to detect rare transcripts using exon bins.

Tool	Major promoters	Minor promoters	Inactive promoters
proActiv	15,498	5,384	69,282
Salmon	27,025	12,657	50,482
DEXSeq	29,198	11,709	49,257

Table 1. Comparison of the three tools in terms of number of promoters found in each group (Major, Minor and Inactive)

Our findings were compared against the benchmarking numbers provided by the SnakeAltPromoter study in the initial research paper. Although we detected 90,164 promoters

by applying the GENCODE hg38-v46 annotation, the research paper noted that there were 118,403 promoters altogether. It is expected that the difference may have arisen due to different GENCODE version numbers used or different filtering criteria set during the Genome Setup process. But, the trends in terms of distributions are identical; specifically, the Major type constitutes (80%) of the total at position 1, whereas the Minor type represents approximately 40% at position 5`.

DATASET 2 (GSE124535)

Applying the SnakeAltPromoter pipeline to the heart failure gene expression data set (GSE124535), we discovered 71,853 promoters in the hg38-v44 genome.

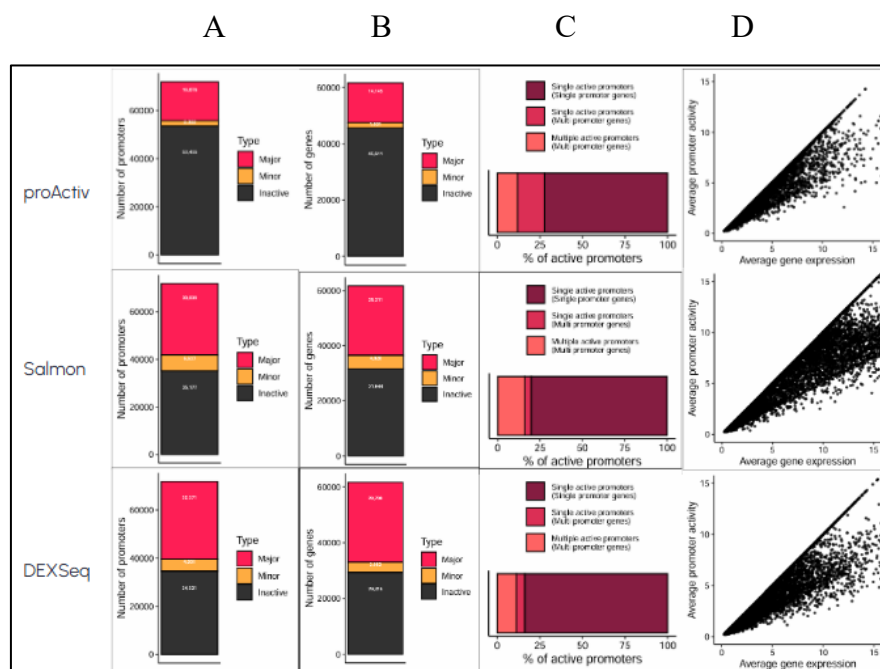


Figure 3. Comparison of tools in GSE124535 A) Classification of promoters into Major, Minor and Inactive. B) Classification of genes into Major, Minor and Inactive. C) Classification of active promoters into Single active and Multi active promoters. D) Correlation of gene expression with average major promoter activity.

The promoter types were classified into three groups: Major, Minor and Inactive. In line with the observations made based on the cardiovascular dataset, cross-comparison within the HCC group exposed a significant level of algorithmic disagreement in the way that short-read information is assembled into separate transcript forms. The proActiv software predicted a realistic promoter profile, where 16,076 promoters were classified as Major, and 2,322 were identified as Minor, with the rest being Inactive promoters at 53,455. Conversely, the quasi-

mapping approach by Salmon resulted in 6,637 promoters that were considered Minor and 35,177 inactive promoters, compared to 30,039 classified as Major promoters. The DEXSeq method, which follows an exon binning technique, showed that 32,371 promoters were classified as Major and 34,531 as Inactive.

Tool	Major promoters	Minor promoters	Inactive promoters
proActiv	16,076	2,322	53,455
Salmon	30,039	6,637	35,177
DEXSeq	32,371	4,951	34,531

Table 2. Classification of promoters among the three tools into Major, Minor and Inactive.

DIFFERENTIAL REGULATION

We performed a systematic identification of Differentially Expressed Genes (DEGs) and Differentially Regulated Promoters (DRPs).

- Heart Failure (Dataset 1): After filtering (cpm rowsum (> 1), non-internal promoters, median activity (> 0.25)), we identified 13,462 genes and their corresponding promoters.

Similarly, we filtered promoters based on external promoters, multi-promoter genes and promoters having median promoter activity more than 0.25 to find DRPs.

- proActiv: Found 51 Upregulated and 102 Downregulated DRPs.
 - Salmon: Found 158 Upregulated and 173 Downregulated DRPs.
 - DEXSeq: Found 101 Upregulated and 246 Downregulated DRPs.
- HCC vs Non-tumor (Dataset 2): After filtering (cpm rowsum (> 1), non-internal promoters, median activity (> 0.25)), we identified 13,470 genes and their corresponding promoters.

We then filtered the promoters that were not internal, promoters that belong to multi-promoter genes and those having median promoter activity more than 0.25.

- proActiv: Found 124 Upregulated and 177 Downregulated DRPs.
 - Salmon: Found 128 Upregulated and 193 Downregulated DRPs.
 - DEXSeq: Found 124 Upregulated and 220 Downregulated DRPs.

DOWNSTREAM ANALYSIS OF AAP

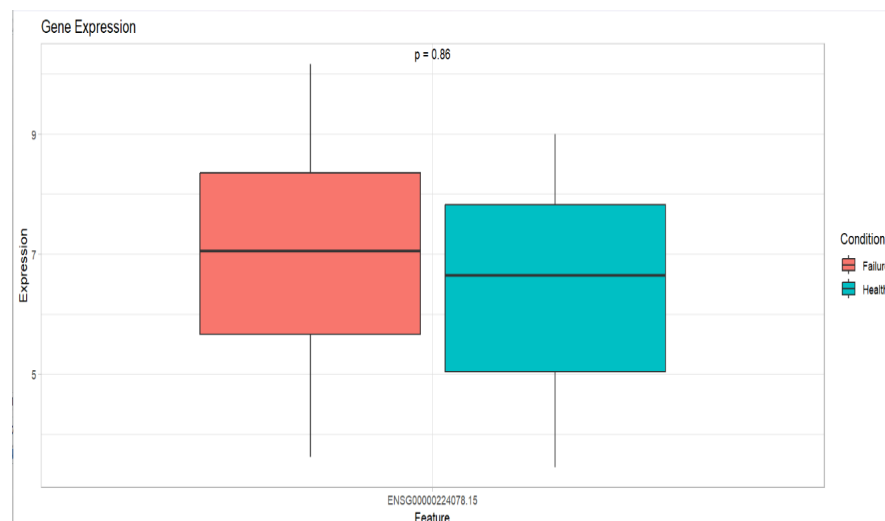
Boxplots were generated to showcase the gene expression and absolute promoter activity of those genes that were non-significant but had both upregulated and downregulated DRPs. The following genes showed interesting results:

Dataset 1:

1. ENSG00000224078.15

The global gene expression showed no statistically significant difference between the heart failure and the healthy control groups ($p = 0.86$). In the context of the conventional bioinformatics approach, which solely depends on the total count of transcripts, the gene ENSG00000224078.15 would be classified as a transcriptionally inactive gene during cardiac remodeling. Nonetheless, in terms of high-definition analysis of its internal promoter structures with proActiv, it turned out that the gene is under dynamic regulation. Absolute promoter activity analysis indicated that although the transcription level of most of its promoters (such as pmtr.62333, pmtr.62334, and many others) were relatively constant, the usage of one particular promoter (pmtr.62388) was statistically different ($p = 0.044$). In fact, the median absolute activity of this promoter showed a marked increase during the failing heart state compared to the healthy state. Another pmtr.62338 seemed to be downregulated in the failing heart state than the healthy state but didn't achieve statistical significance ($p = 0.359$), presumably due to limited samples. In a research done by Feng et al. (2025), this gene, also known as the SNHG14 gene, has shown to be a potential target for cancer therapy as it is a promising biomarker for the identification of tumors.

(a)



(b)

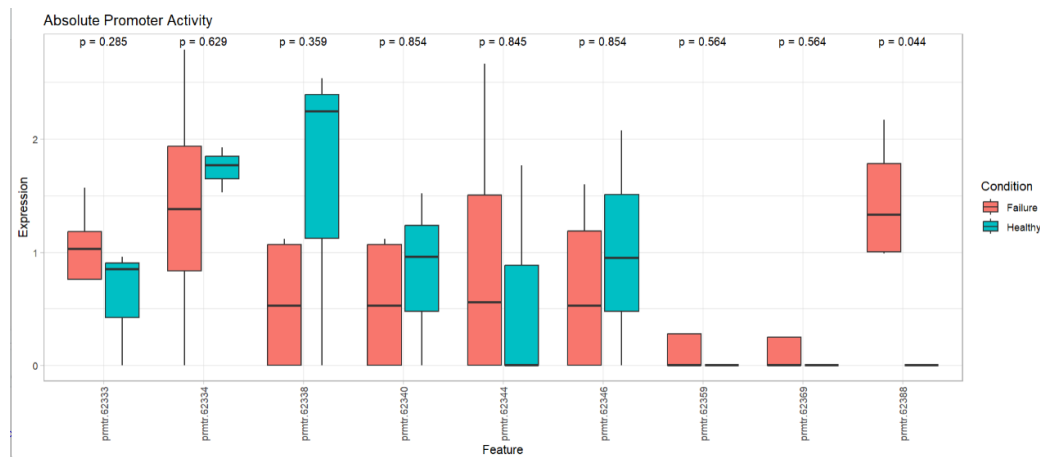


Figure 4. (a) Gene expression boxplot of gene ENSG00000224078.15. (b) Absolute promoter activity of gene ENSG00000224078.15 depicted.

2. ENSG00000143515.20

Based solely on the total RNA output from this gene, there seems to be no problem. The boxplot showing the comparison between "Failure" and "Healthy" conditions for this gene reveals that the output between the two groups is statistically the same. The analysis produced an extremely non-significant p-value of 0.86. So, at the gene level, this particular gene looks completely unaffected by heart failure. The biological significance of this gene changed once we analysed the absolute promoter activity of the start sites of the gene through our pipeline. It turns out that this particular gene is regulated by three different promoters, and a failing heart was actively changing them in the following manner- pmtr.27986 is being actively used in healthy hearts, but the same promoter is silenced in heart failure samples. The p-value produced by our analysis equals 0.044. While the heart shuts down, the activity is shifting from promoter.27986 pmtr.27984. However, while p-value for promoter.27984 equals 0.057, we have to treat this finding as a trend rather than a statistically significant observation (since p-values greater than 0.05 do not qualify).

(a)

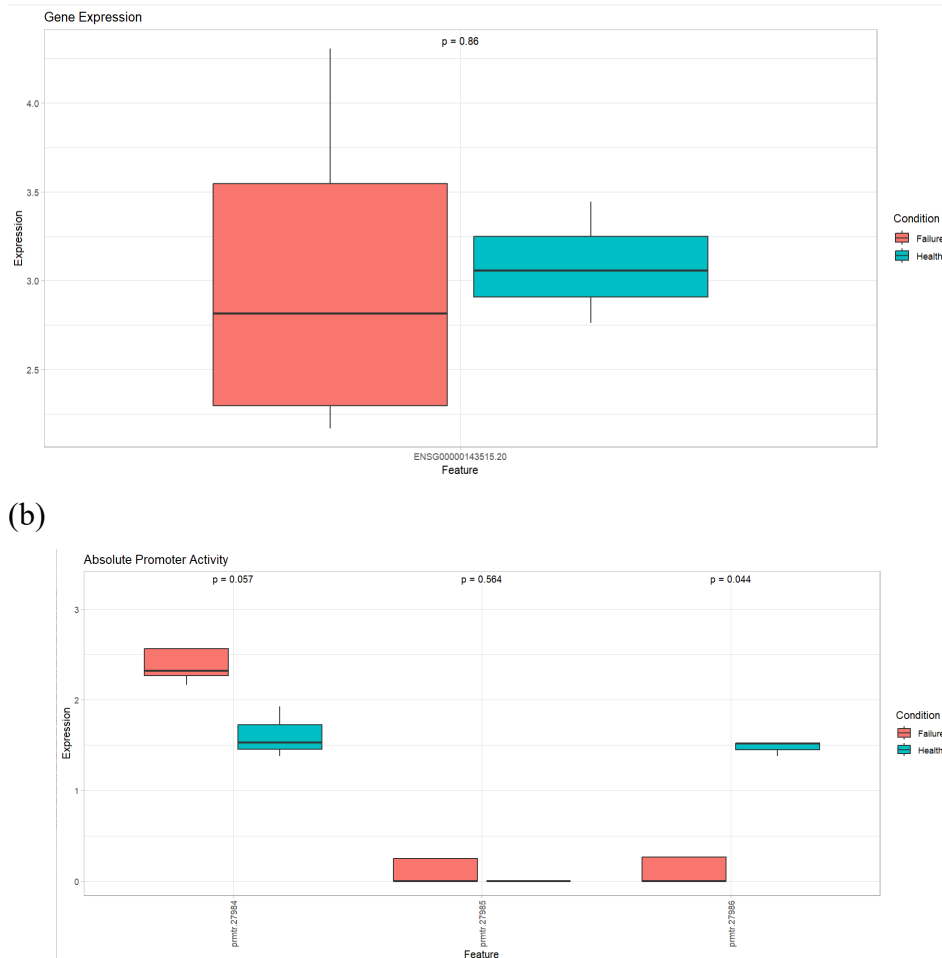


Figure 5. (a) Gene expression boxplot of gene ENSG00000143515.20. (b) Absolute promoter activity of gene ENSG00000143515.20 depicted.

Dataset 2:

Some genes taken from proActiv result:

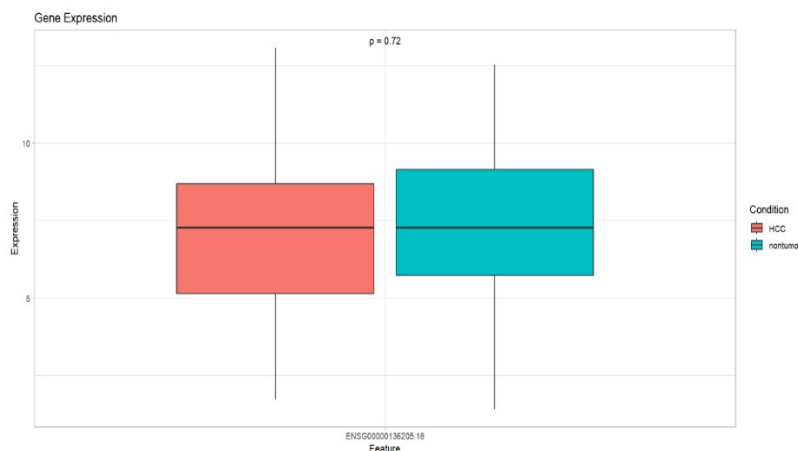
1. ENSG00000136205.18

While analyzing total summed transcript values, statistical analysis showed that there is no absolute difference between the transcripts present in the Hepatocellular Carcinoma and the non-cancerous tissue adjacent to it (p-value = 0.72). In terms of standard gene expression quantification, this particular gene shows no changes when comparing the two samples mentioned above. However, after breaking down the gene into individual promoters, the results reveal that there is severe internal disruption. While some promoters of the alternative promoters remain biologically inactive or are not changed (pmtr.12064, p-value = 0.988; pmtr.12068, p-value = 0.836), certain nodes show drastic changes in their expression levels. Specifically, the absolute promoter value of pmtr.12063 shows highly significant usage (p-value = 7.3×10^{-6}).

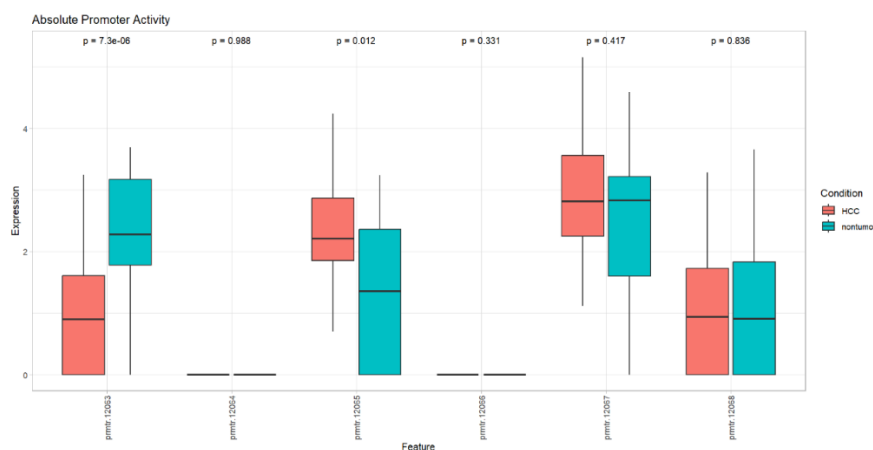
Another prmr.12065 showed significant downregulation ($p = 0.012$) in healthy samples. The stratification of patients according to the expression of prmr.12063 showed a significant difference in Disease-Free Survival ($p = 0.039$). The group with high expression of this particular promoter variant ($n=20$) displayed clear differences after treatment from the low expression group ($n=15$). In contrast, even though there was differential expression between the healthy and tumor tissues of prmr.12065, the difference in patient survival due to expression of this promoter was insignificant ($p = 0.21$). These results emphasize the need for an incredibly high level of specificity when looking for biomarkers and the stratification of patients with prmr.12063 provides evidence alternatively spliced promoters can be effective biomarkers. Isolating the particular isoform responsible for poor prognosis provides the kind of biological resolution impossible at the gene level.

This gene, also known as TNS3, hasn't been associated with liver cancer yet but has been a tumor promoter for other cancers like lung cancer (Shen et al., 2021).

(a)



(b)



(c)

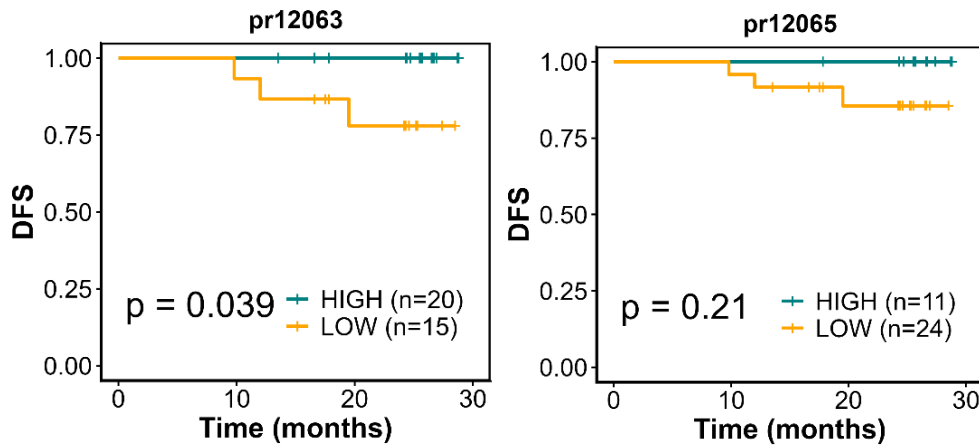
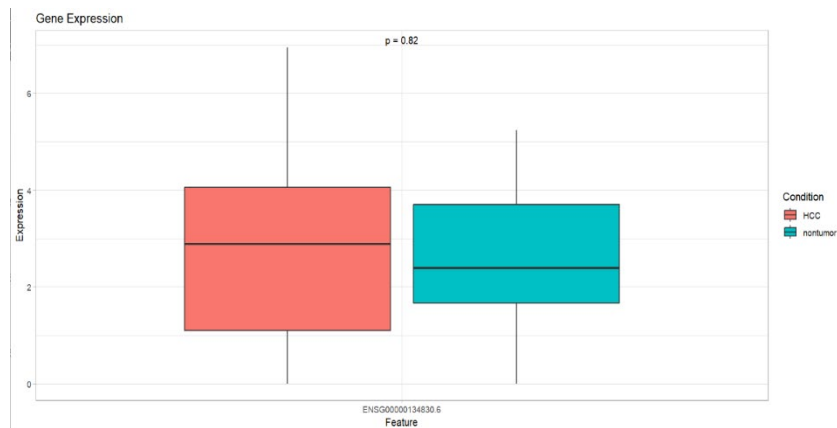


Figure 5. (a) Gene expression boxplot of gene ENSG00000136205.18. (b) Absolute promoter activity of gene ENSG00000136205.18 depicted. (c) Disease-free survival analysis of promoters 12063 and 12065.

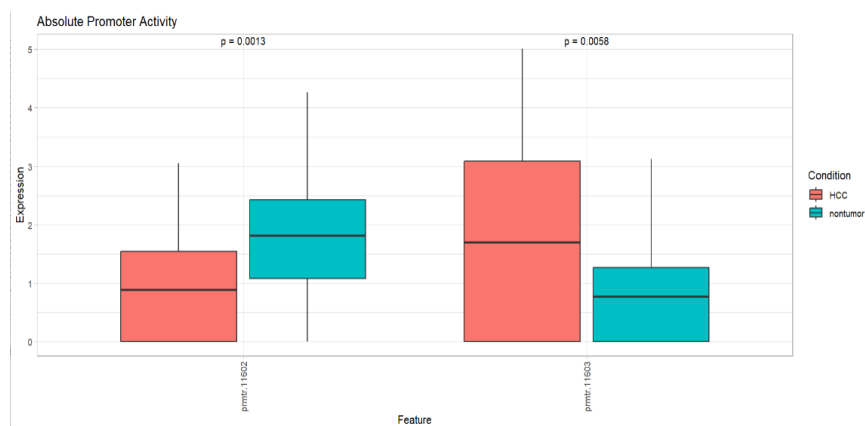
2. ENSG00000134830.6

As seen from the previous results, no dysregulation of the gene could be found using conventional quantitation methods, leading to an insignificant differential expression profile between HCC samples and non-tumor groups ($p = 0.82$). However, through isoform-specific analysis using proActiv, a very dynamic internal regulation pattern of the gene was observed. In this particular case, the two alternatively regulated promoters showed significant differences: the alteration of pmtr.11602 was highly significant ($p = 0.0013$), while pmtr.11603 also displayed a very significant differential expression pattern ($p = 0.0058$). The clinical applicability of such changes, however, emphasizes the strict criteria needed for biomarker identification. Although the alternative promoters have definitely been affected by the transformation process, Kaplan-Meier survival analysis showed that they are unable to serve as reliable independent markers. Patients divided according to the absolute level of expression of promoter.11602 could not be reliably differentiated with regard to DFS curves ($p = 0.12$), whereas the expression of promoter.11603 was close but not significant enough to achieve stratification ($p = 0.078$). This gene is called C5aR2 and it has been known to be significantly upregulated in HCC tissues according to a study by Qian et al., 2021. High expression of this gene is associated with poor survival rates.

(a)



(b)



(c)

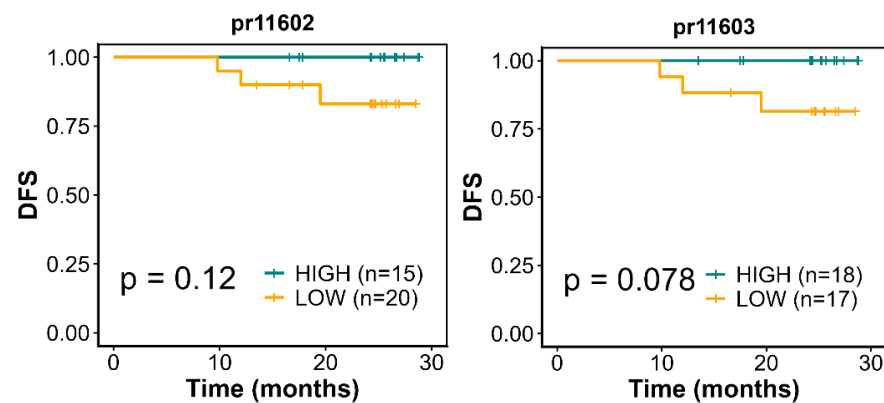


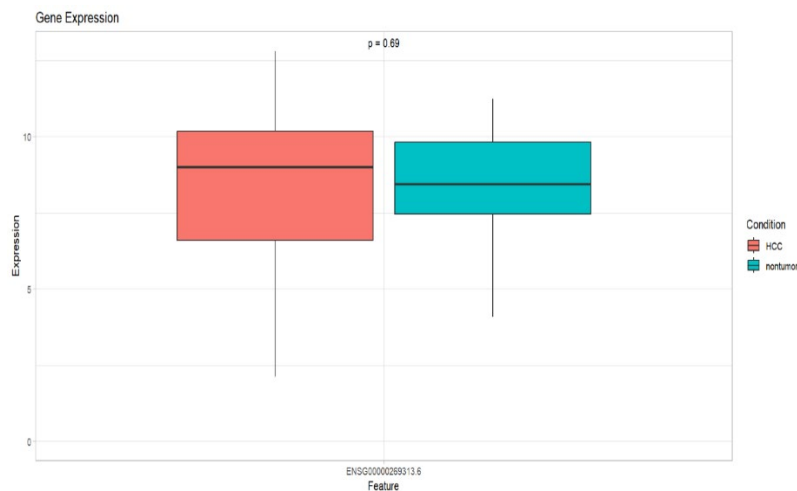
Figure 6. (a) Gene expression boxplot of gene ENSG00000134830.6. (b) Absolute promoter activity of gene ENSG00000134830.6 depicted. (c) Disease-free survival analysis of promoters 11602 and 11603.

3. ENSG00000269313.6

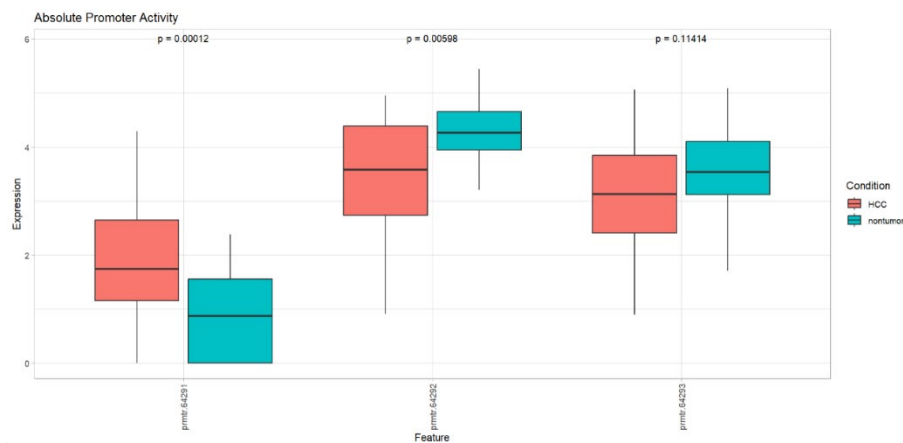
Despite the global gene expression analysis showing that there is no difference in the overall expression level of the ENSG00000269313.6 transcripts during

hepatocarcinogenesis (HCC vs non-tumor: $p = 0.69$), isoform deconvolution using an automated pipeline showed significant changes in the structure. Specifically, the activity level of the pmtr.64291 promoter is dramatically changed in the tumor compared to the non-tumor sample ($p = 0.00012$), and this is one of the most statistically significant promoter changes in the set. Another promoter at the same locus, namely pmtr.64292, is significantly affected ($p = 0.00598$). Importantly, the local changes in transcriptional programming are not simply a molecular phenomenon; they result in an exceedingly aggressive phenotype. Kaplan-Meier survival analysis indicated that the actual activity of pmtr.64291 constitutes an extremely potent biomarker in terms of prognosis. Patients divided based on the high level of the given alternative promoter showed a drastic change in their Disease-Free Survival profile as compared to those in the low category ($p = 0.008$). On the other hand, the second promoter, pmtr.64292, failed to show any independent prognostic value ($p = 0.088$).

(a)



(b)



(c)

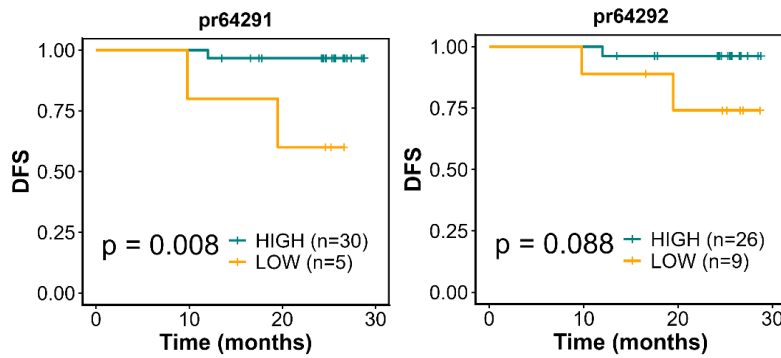


Figure 7. (a) Gene expression boxplot of gene ENSG00000269313.6. (b) Absolute promoter activity of gene ENSG00000269313.6 depicted. (c) Disease-free survival analysis of promoters 64291 and 64292.

The pmtr.64291 marker provides evidence for the primary thesis statement, which postulates that important oncogenic driver genes and clinical biomarkers are often concealed by the isoform structure of genes that are transcriptionally stable.

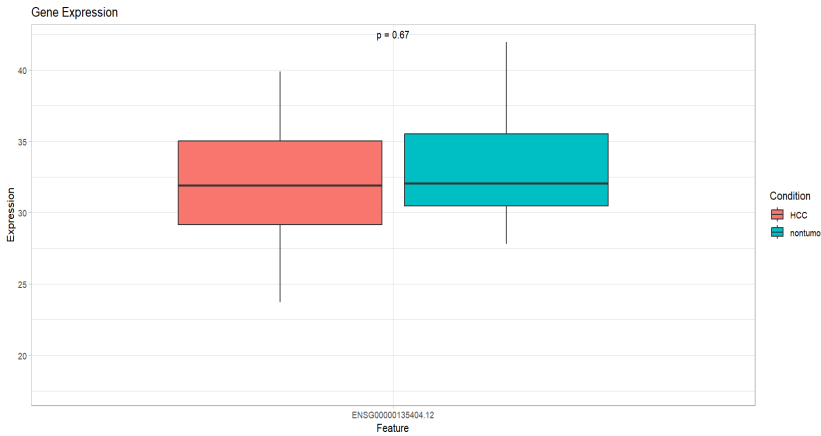
Some genes taken from Salmon result:

1. ENSG00000135404.12

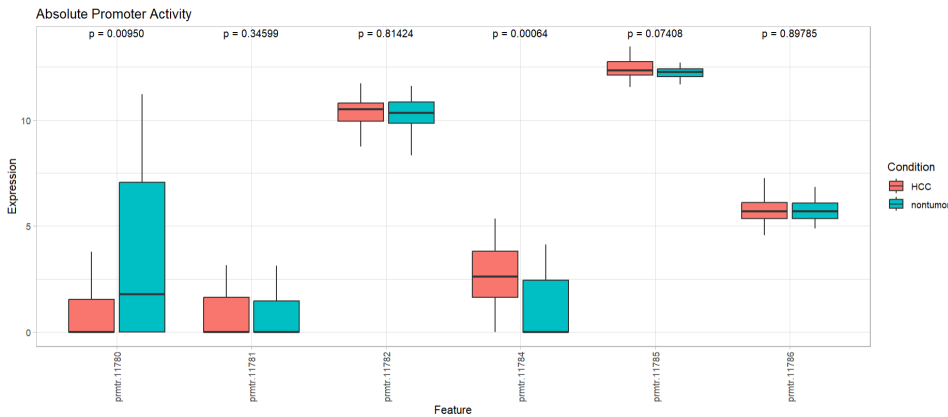
This particular gene constitutes a classic example of a "promoter switch," whereby the tumor drastically modifies the gene's interpretation while keeping the level of RNA the same. The initial assessment of the conventional, whole-gene expression data on ENSG00000135404.14 was rather dull. When we compared the distribution between the HCC (tumor) samples and non-tumor control samples by a boxplot, it showed an almost identical distribution, with p-value=0.67. If a researcher was solely relying on conventional gene expression data in their research, they would easily come to the conclusion that the gene is not involved in hepatocellular carcinoma. But when we looked at the promoter level (as can be seen in figure 10.), we could see that pmtr.11780 was significantly downregulated in tumor tissues as compared to healthy tissue ($p = 0.00950$) while another pmtr.11784 was significantly upregulated in tumor tissues as compared to healthy tissues ($p = 0.00064$). This clearly depicts alternative promoter switching. The process involves turning off one normal promoter and turning on a cancer-specific promoter, and effectively alters the final transcript isoforms of the RNA while producing the exact total amount of gene expression. It is important to establish whether such alterations affect the prognosis for the patient. For this purpose,

we conducted two Kaplan-Meier survival tests using the two dysregulated promoters and attempted to find any correlation between the activity level of each of them and DFS over the course of 30 months. Despite the fact that pmtr.11780 has shown considerable suppression in cancer patients, its presence in the body proved to have no effect on patient prognosis. As shown from our tests, the survival functions for patients with high activity (n=10) compared with those with low activity (n=25) of this promoter do not demonstrate any statistically significant difference ($p = 0.26$). However, the other pmtr.11784, could be a useful biomarker in clinical settings. High expression level (n=15) of pmtr.11784 was associated with significantly poor disease-free survival rate, specifically, the survival curve for patients with high pmtr.11784 expression dropped dramatically, whereas patients with low levels (n=20) showed no disease progression. The difference between both groups was statistically significant at $p=0.045$. This gene, also known as CD63, has been proven to be associated with tumor progression in hepatocellular carcinoma so it can be used as a potential target for cancer therapies (Liu et al., 2024).

(a)



(b)



(c)

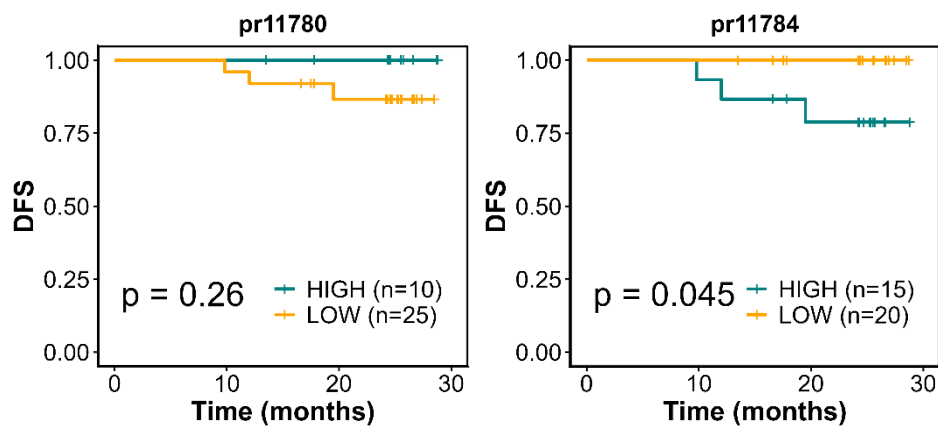


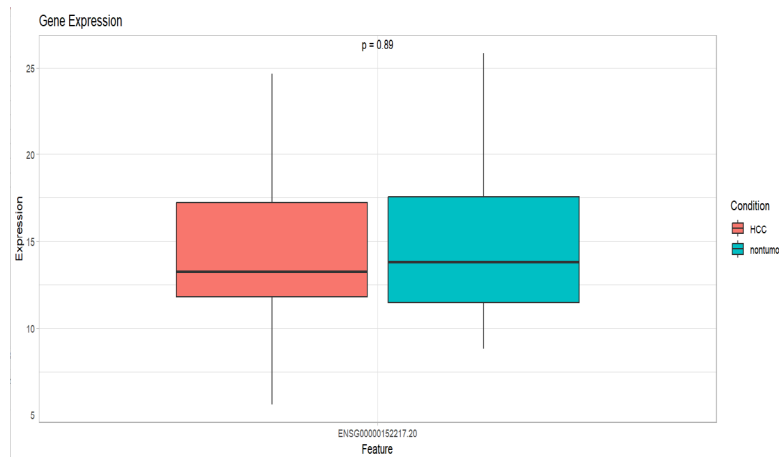
Figure 8. (a) Gene expression boxplot of gene ENSG00000135404.12. (b) Absolute promoter activity of gene ENSG00000135404.12 depicted. (c) Disease-free survival analysis of promoters 11780 and 11784.

2. ENSG00000152217.20

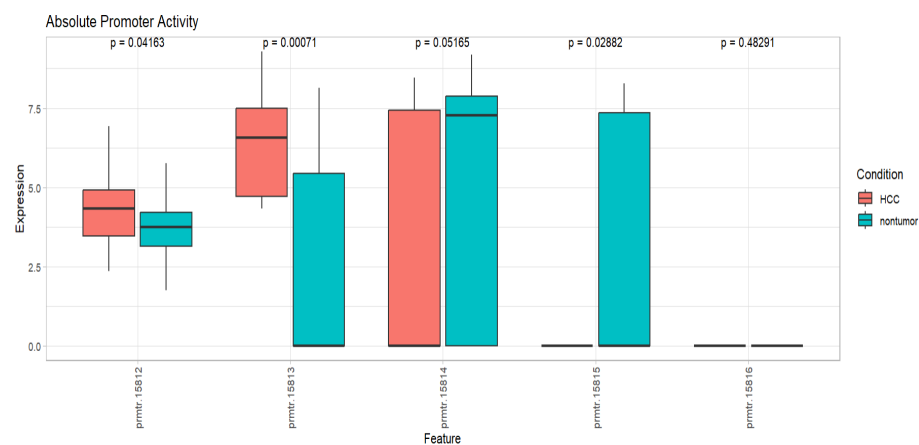
This particular gene shows very well how tumors have multiple potential changes but only selected ones pose threats to patients. Boxplot comparison of the gene's expression levels in the Hepatocellular Carcinoma (HCC) group versus normal non-tumor group was nearly identical. The p-value of the statistics test conducted was highly non-significant at 0.54. Using traditional bioinformatics approaches based on quantification of total RNA production, we would have not considered this gene in our study because we would have assumed it would not be influenced by the cancer cells. However, when focusing on absolute activity of each promoter in the gene, we discovered that this gene is controlled by more than one promoter and the tumor interferes with their activity. We can see that pmtr.15813 was significantly upregulated in tumor tissues as compared to healthy tissues ($p = 0.00071$) while pmtr.15815 showed significant upregulation in healthy tissues and no activity in tumor tissues ($p = 0.02882$). Other promoters, like pmtr.15814 ($p = 0.05165$) and pmtr.15816 ($p = 0.48291$), didn't show significant change in HCC and tumor tissues. We conducted Kaplan-Meier survival analysis to find out whether the expression of the promoters that have become activated due to this mutation can serve as a predictor of Disease-Free Survival (DFS) time for the patient. Although the tumor strongly induced the expression of pmtr.15813, this alteration does not seem to be associated with a statistically significant increase in the

DFS. On comparing the survival times for patients with high levels of the gene expression (n=4) and low levels (n=31), we found that the p-value was 0.16. Since this value is greater than 0.05, we cannot assume any statistically significant differences between the two groups' survival rates. The same trend was observed for the other promoter. For patients with high levels of expression (n=7) vs low levels (n=28), the corresponding p-value was 0.34. This gene, also known as SETBP1, has been linked with various cancers and Schinzel-Giedion syndrome. However, it has not been specifically associated with HCC but it can still be explored as a potential biomarker (Kohyanagi et al., 2023).

(a)



(b)



(c)

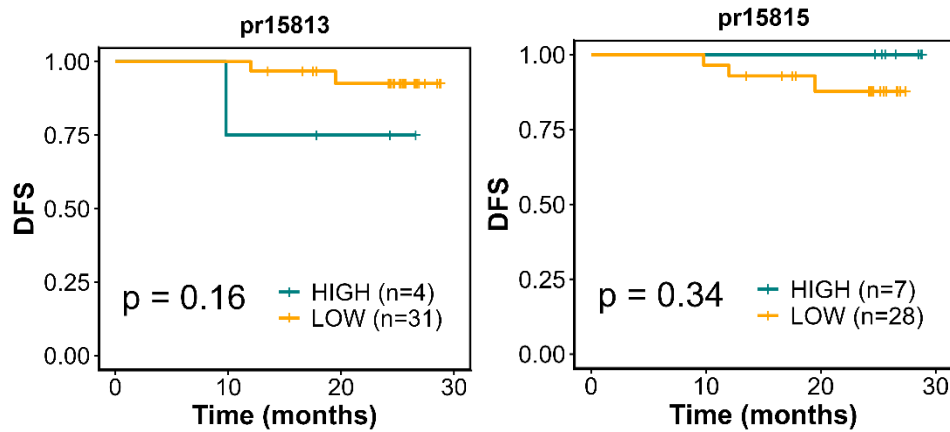


Figure 9. (a) Gene expression boxplot of gene ENSG00000152217.20. (b) Absolute promoter activity of gene ENSG00000152217.20 depicted. (c) Disease-free survival analysis of promoters 15813 and 15815.

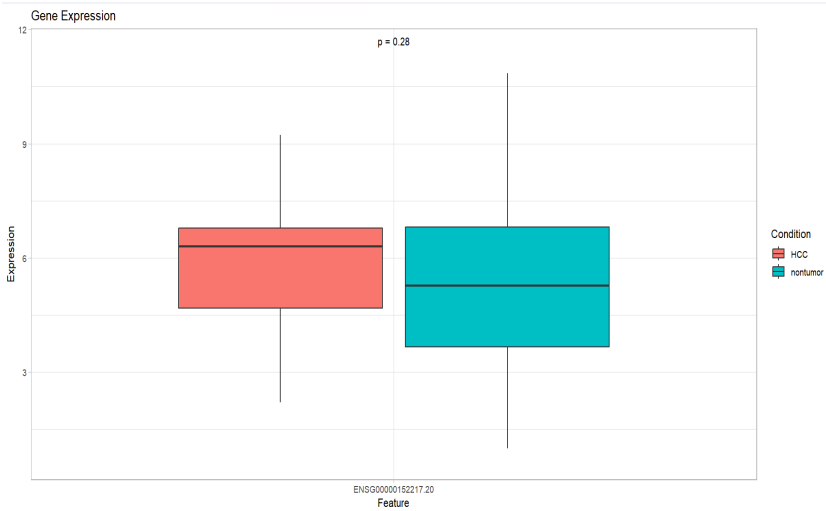
Some genes taken from DEXSeq result:

1. ENSG00000152217.20

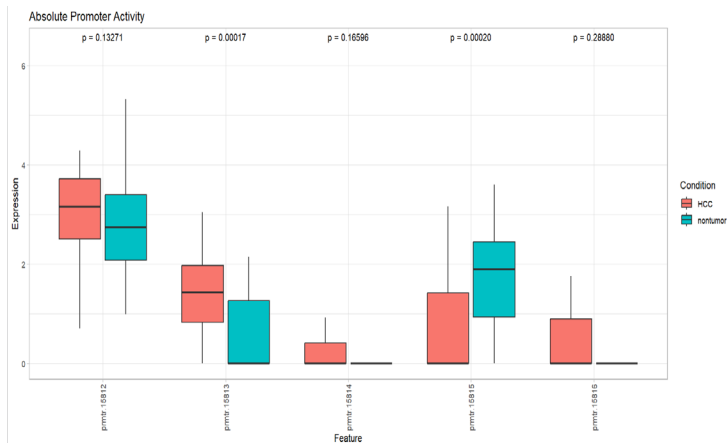
When looking at the boxplot for gene expression level, we could see that there is no significant change ($p = 0.28$) in gene output so it appears that nothing is affecting the gene. But when we looked at the promoter level, we found a very significant shift in the activities of two promoters. As we can see, `prmr.15813` shows significant upregulation ($p = 0.00017$) in tumor tissues while `prmr.15815` is significantly downregulated ($p = 0.00020$) in tumor tissues. A similar trend is shown for the same gene that was evaluated using the tool Salmon's result. However, when using the result obtained from DEXSeq, it shows more significance value as compared to Salmon. Then, to see how this gene impacts a patient's prognosis, we performed Kaplan-Meier survival analysis on the significant promoters i.e. `prmr.15813` and `prmr.15815`. This tumor had greatly increased the `prmr.15813` expression levels, and this is evident from the survival data, which shows that there is a strong relationship between the two factors. When the small number of patients with high expression level ($n=4$) were compared with those with low expression level ($n=31$), the p-value was 0.00068, showing a significant difference between the survival curves. The survival analysis of `prmr.15815` however revealed no statistically significant separation between the two groups. The log-rank test returned a p-value of 0.2. As this is well above the 0.05

threshold, it implies that the downregulation of pr15815 is likely just a side-effect of the chaotic tumor environment, rather than a driver of HCC. So, this gene can be a potential target for cancer treatments.

(a)



(b)



(c)

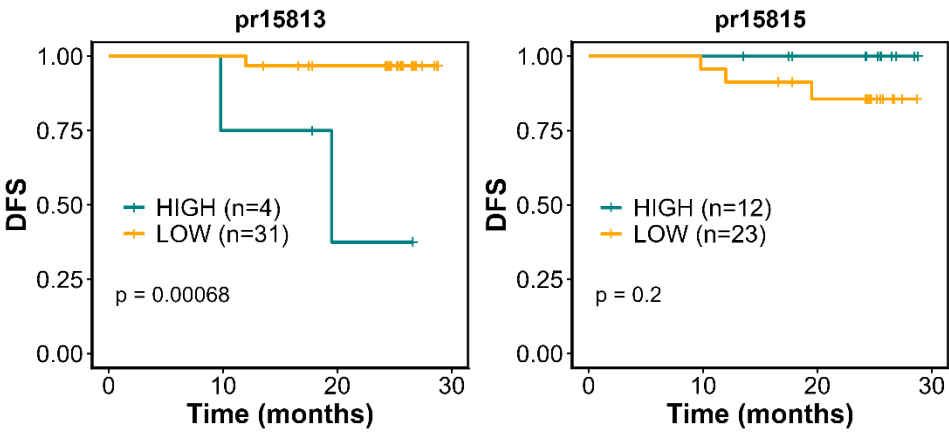


Figure 9. (a) Gene expression boxplot of gene ENSG00000152217.20 from DEXSeq results. (b) Absolute promoter activity of gene ENSG00000152217.20 depicted. (c) Disease-free survival analysis of promoters 15813 and 15815.

CHIP-SEQ DATA ANALYSIS

We used publicly available ChIP-seq data (**GSE276132**) for H3K4me3 and H3K27ac in PLC cell line for HCC to validate our findings. The track plot shows promoters of SETBP1 gene corresponding to the H3K4me3 and H3K27ac peaks.

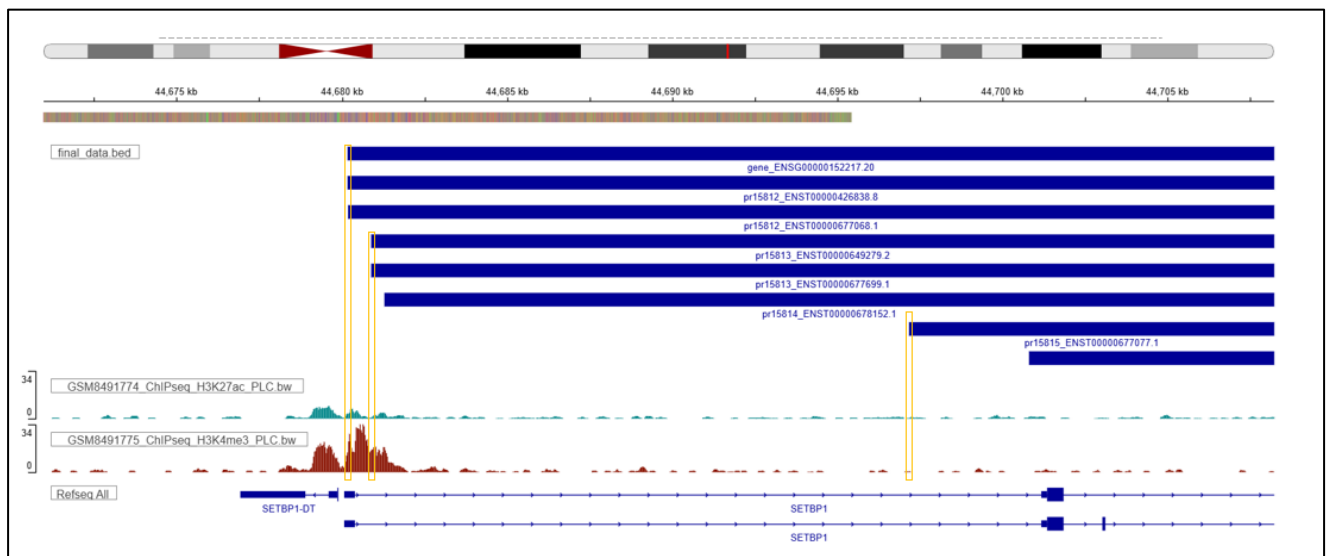


Figure 10. Track plot for promoters of gene SETBP1 corresponding to H3K4me3 and H3K27ac peaks.

Conclusions

The current study demonstrates the significance of using the SnakeAltPromoter algorithm in understanding the complexity and overlap in transcriptomic organization from regular RNA sequence data. This research has shown that the use of alternative promoters can be used as an additional regulatory level to differentiate between normal and pathological human physiology by applying the automatic system on patients suffering from heart failure. We also tested the pipeline on tumor and non-tumor samples of patients with Hepatocellular Carcinoma.

However, before we could confidently use the biological results, we had to overcome a fundamental computational challenge: there was no consensus among several bioinformatics

tools used to interpret the sequencing data. While comparing Salmon, DEXSeq, and proActiv, we discovered a big difference in how they estimated active promoters. Sequencing tools such as Salmon had difficulty dealing with ambiguous short reads, thus significantly overcounting the number of active minor promoters. On the contrary, proActiv yielded far more reliable results by focusing on junction reads connecting the first two exons of a gene. Hence, in modern bioinformatics, choosing a particular algorithm is crucial for interpreting the biological findings. In both the cardiovascular and cancer datasets, we observed several genes having normal gene expression profiles. Traditional RNA-sequencing would have ignored these genes altogether, assuming their expression to be healthy. However, upon further investigation, we discovered evidence of chaotic rearrangements inside the cell. Tumor cells actively turned off the healthy promoter regions and activated cancer-specific ones. Thus, our findings prove that malignant cells could significantly alter protein production without changing the overall intensity of expression of the corresponding gene, making the diagnosis difficult. Certain genes identified during the process, like TNS3, CD63, SETBP1, have been mentioned in literature as tumor progressors so these results are validated.

There are certain limitations also that were found during the study. The whole research process utilizes short-read RNA sequencing alone. Since short-read sequencing requires RNA to be broken down into small pieces, identifying the exact start promoter for each fragment is based on complex calculations. Even though proActiv successfully calculates it, the data obtained cannot be considered flawless. In particular, when analyzing our results through survival analysis, we had a very uneven distribution within some groups, as one group had only four patients compared to 31 patients in another group. Despite the fact that our calculations showed significance anyway, small sample size is always associated with potential bias. Larger samples will allow us to be confident about those biomarkers. This research was completely computational. Though mathematics and statistics have allowed us to conclude that these alternate promoters are responsible for cancer development, we did not conduct wet-lab research to validate these results.

However, with these limitations also, several future research paths can be explored. An immediate future action would be repeating the above analysis using the new generation Long Read sequencing methods (e.g., Oxford Nanopore, PacBio). This method sequences the entire RNA from the beginning to the end without any gaps, thus leaving no doubts about which promoter is responsible for the alternative transcription. During this research, it was proven that SnakeAltPromoter can be effectively utilized for Hepatocellular carcinoma datasets. Further studies should include analysis of RNA in other cancers like breast cancer, lung cancer,

and brain cancer in order to determine whether there are promoters that are common for all types of cancer or organ-specific. In order to achieve personalized medicine, it is crucial to understand the nature of cancer at a molecular level. Thus, once it is known that a particular promoter leads to cancer recurrence and not the entire gene, drug developers can target that particular transcript leaving the rest of the gene intact.

To conclude, this research demonstrates the efficiency of the SnakeAltPromoter pipeline in decoding intricate transcriptomic information. The importance of exploring only the overall gene expression level is insufficient to provide the true mechanism underlying various diseases in humans. Exploring more deeply at the isoform level allowed us to reveal how failing hearts and cancerous liver tumors secretly alter their genomic instructions.

Most importantly, by connecting such transcriptomic alterations with the prognosis of patients, the current research confirms the key role of alternative promoters as major regulators, strong biomarkers, and excellent therapeutic targets.

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